

SAI CHARAN REDDY KOTHAKAPU

Data Analyst | Software Engineer

Utica, New York

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SUMMARY

Software Engineer and Data Analyst with hands-on experience building, supporting, and improving data-driven and backend systems in a regulated public-sector environment. Strong foundation in Python, SQL, APIs, and databases, with experience developing dashboards, automating workflows, and translating raw data into clear, actionable insights. Comfortable working across the full problem lifecycle, from requirements gathering and system analysis to debugging, documentation, and collaboration with technical and non-technical stakeholders to deliver reliable and scalable solutions.

SKILLS

Analytics & Data: KPI Tracking, Data Cleaning, Data Validation, Ad-hoc Analysis, Dashboard Development, Reporting Automation, Metadata Management, Semantic Search, Basic Experiment Analysis (A/B Testing)

SQL & Databases: SQL (Joins, Subqueries, CTEs, Window Functions), Query Optimization, Relational Databases, Vector Databases, Data Modeling Basics (Fact & Dimension Tables)

BI & Visualization: Power BI, Excel, Google Sheets, Pandas, NumPy, Matplotlib

Programming & Backend: Python, FastAPI, REST APIs, Workflow Automation, Debugging, Git, Unit Testing Basics (pytest)

Systems & Design: Linux, Windows, Cloud Fundamentals (AWS, Azure), System Monitoring & Logging, User Access Management, System Design Basics

Tools & Collaboration: Jira, Confluence, Technical Documentation, Requirements Gathering, SDLC Support, Stakeholder Reporting, Code Reviews

EXPERIENCE

Graduate Assistant | Data & Systems Support

Sep 2024 - Dec 2025

Health Sciences Department, SUNY Polytechnic Institute

- Built and automated Excel- and SQL-based tracking workflows, reducing manual clerical effort by 8+ hours per week and improving reporting reliability.
 - Developed Power BI dashboards to support equipment planning, budgeting, and operational decision making for department stakeholders.
 - Performed data validation, record review, and correction across datasets, reducing documentation and filing errors by 30%.
 - Translated raw operational data into clear, decision-ready summaries to support ad-hoc reporting and analysis.
 - Troubleshooted and supported enterprise applications by debugging workflows, resolving user issues, and maintaining reliable system operations.
 - Worked with databases and backend systems to support data pipelines, ensure record consistency, and maintain data integrity.
 - Participated in system lifecycle activities including requirements gathering, testing support, and technical documentation.
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PROJECTS

Professor Agent - Research-Enhanced Knowledge Automation System (Academic Capstone Project)

- Designed and built an AI-driven system to automate research content summarization and knowledge retrieval, reducing manual lecture update time by 70%+.
- Implemented a vector-based semantic search pipeline to process and index 100+ PDF, DOCX, and PPTX documents, enabling fast, context-aware retrieval.
- Integrated external data sources via APIs (arXiv, YouTube) and built validation and compliance checks including citation verification, URL validation, hallucination detection, and audit logging.
- Developed structured data pipelines to store, index, and retrieve content reliably, improving accuracy and trust in generated outputs.

Multi-Agent Task Automation System (Independent Automation Project)

- Built a multi-agent automation platform that converted natural-language requests into coordinated

workflows across messaging, scheduling, file search, and web services.

- Designed backend workflows to orchestrate 5+ service agents, enabling automated reminders, task tracking, and document coordination.
- Supported academic operations for 40+ students, reducing manual administrative effort by 60%+ through automation.
- Integrated multiple services and APIs to ensure reliable task execution, error handling, and system coordination.

EDUCATION

SUNY Polytechnic Institute

Master's in Computer Science

Utica, New York

Jan 2024 - Dec 2025

Malla Reddy Institute of Engineering & Technology

Bachelor of Technology in Computer Science and Engineering

Hyderabad, India

2019-2023